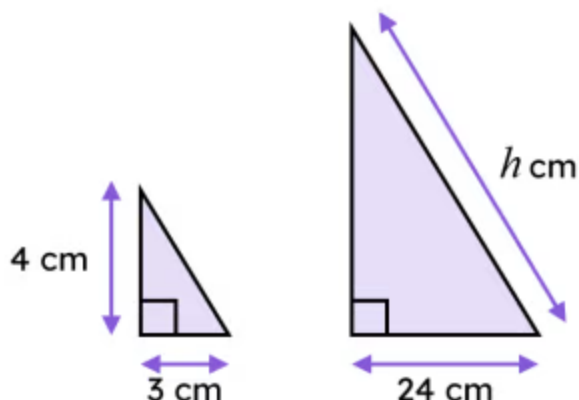
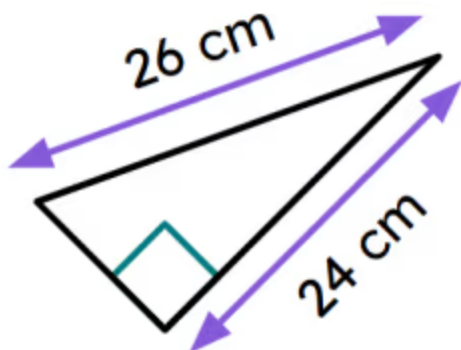


Problem solving with similarity and Pythagoras' theorem

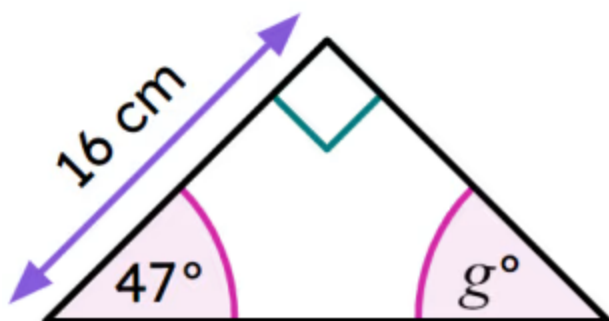
- 1 These two triangles are similar to each other and are both in the same orientation. The length of the hypotenuse marked h is 40 cm. Fill in the blank



- 2 The area of this triangle is 120 cm². Fill in the blank

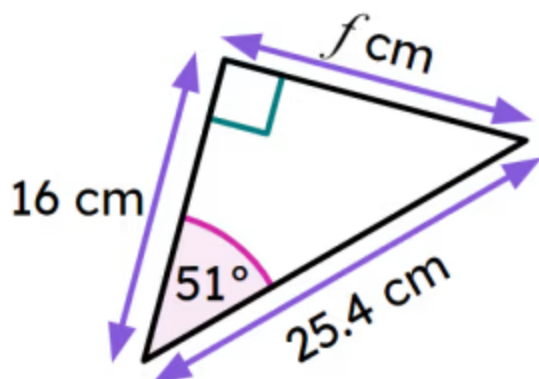


- 3 Which of these statements are correct for this triangle? Tick **2** correct answers



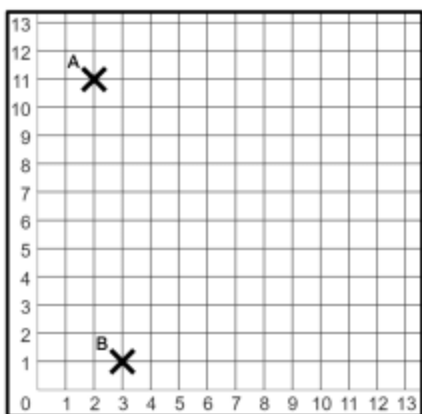
- ☐ You need to use Pythagoras' theorem to find the size of angle g .
- ☒ You do not need to use Pythagoras' theorem to find the size of angle g .
- ☐ $g^\circ = 27^\circ$
- ☒ $g^\circ = 43^\circ$
- ☐ $g^\circ = 47^\circ$

4 Which of these statements are correct for this triangle? Tick **2** correct answers



- ☒ The perimeter of this triangle is $16 + 25.4 + 19.7 = 61.1$ cm (to 1 d.p.).
- ☐ The perimeter of this triangle is $16 + 25.4 + 25.4 = 92.4$ cm (to 1 d.p.).
- ☒ The area of this triangle is 158 cm^2 (to the nearest cm^2).
- ☐ The area of this triangle is 203.2 cm^2 (to 1 d.p.).
- ☐ $f^\circ = 180^\circ - 90^\circ - 51^\circ = 39^\circ$.

5 Each square is 1 unit in length. The shortest distance from point A to point B is 10.05 units (rounded to 2 d.p.). Fill in the blank



6 Point C is at the coordinate (12, 3). The perimeter of a triangle whose vertices are at points A, B, and C is 32 units (to the nearest unit). Fill in the blank

